VPN IPSec IKEv2 Site-to-Site Basado en Rutas (VTI)

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Práctica: P4

Objetivo

Configurar un túnel VPN IPSec IKEv2 Site-to-Site basado en rutas mediante una **Virtual Tunnel Interface (VTI)** entre dos routers Cisco CSR1000v, permitiendo la comunicación cifrada entre dos redes LAN a través de un router ISP. Esta práctica combina la sintaxis de IKEv2 (vista en P4) con la arquitectura de túnel VTI (vista en P2), representando la evolución natural de ambas tecnologías hacia una solución más moderna, eficiente y escalable.

La diferencia fundamental respecto a P2 (IKEv1 VTI) es que el `crypto ipsec profile` ahora referencia un `ikev2-profile` explícito en vez de usar IKEv1 implícitamente. El resto de la arquitectura es idéntica: una interfaz Tunnel0 de tipo `ipsec ipv4` enruta todo el tráfico que atraviesa el túnel, cifrándolo automáticamente sin necesidad de ACLs.

Comparación de las 6 prácticas Site-to-Site

Práctica	Protocolo IKE	Método	ACL necesaria	Multicast
P1	IKEv1	Policy-Based	Sí	No
P2	IKEv1	VTI	No	No
P3	IKEv1	GRE over IPSec	Sí (GRE/47)	Sí
P4	IKEv2	Policy-Based	Sí	No
P5	IKEv2	VTI	No	No
P6	IKEv2	GRE over IPSec	Sí (GRE/47)	Sí

Topología

```
[PC1]--[SW-A]--[R1]--------[ISP]--------[R2]--[SW-B]--[PC2]
        LAN-A   Gi3    Gi1       Gi2    Gi2    LAN-B
              25.7.73.137  25.7.73.129  25.7.73.133  25.7.73.134  25.7.73.145

                    Tunnel0 (VTI)          Tunnel0 (VTI)
                 25.7.73.153 <----------> 25.7.73.154
                      IPSec IKEv2 (profile PROF-IPSEC-P5)
```

Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Máscara	Descripción
R1	GigabitEthernet1	25.7.73.129	255.255.255.252	WAN R1 → ISP
ISP	GigabitEthernet1	25.7.73.130	255.255.255.252	WAN ISP → R1
ISP	GigabitEthernet2	25.7.73.133	255.255.255.252	WAN ISP → R2
R2	GigabitEthernet2	25.7.73.134	255.255.255.252	WAN R2 → ISP
R1	GigabitEthernet3	25.7.73.137	255.255.255.248	Gateway LAN-A
PC1	eth0	25.7.73.138	255.255.255.248	Host LAN-A
R2	GigabitEthernet3	25.7.73.145	255.255.255.248	Gateway LAN-B
PC2	eth0	25.7.73.146	255.255.255.248	Host LAN-B
R1	Tunnel0 (VTI)	25.7.73.153	255.255.255.252	Extremo VTI R1
R2	Tunnel0 (VTI)	25.7.73.154	255.255.255.252	Extremo VTI R2

Subredes utilizadas (bloque 25.7.73.128/27)

Subred	Rango utilizable	Uso
25.7.73.128/30	25.7.73.129 – 25.7.73.130	WAN R1 – ISP
25.7.73.132/30	25.7.73.133 – 25.7.73.134	WAN ISP – R2
25.7.73.136/29	25.7.73.137 – 25.7.73.142	LAN-A
25.7.73.144/29	25.7.73.145 – 25.7.73.150	LAN-B
25.7.73.152/30	25.7.73.153 – 25.7.73.154	Túnel VTI

Parámetros de configuración IPSec IKEv2

IKEv2 Proposal

Parámetro	Valor
Cifrado	AES-CBC 256
Integridad	SHA-256
Grupo DH	Group 14 (2048-bit)

IKEv2 Policy

Parámetro	Valor
Nombre	POL-P5
Proposal asociado	PROP-P5

IKEv2 Keyring

Parámetro	R1	R2
Peer remoto	R2 (25.7.73.134)	R1 (25.7.73.129)
Pre-shared key	Cisco123!	Cisco123!

IKEv2 Profile

Parámetro	R1	R2
Match identity remote	25.7.73.134/32	25.7.73.129/32
Autenticación local	pre-share	pre-share
Autenticación remota	pre-share	pre-share
Keyring asociado	KEY-P5	KEY-P5

IPSec Profile (diferencia clave vs P2)

Parámetro	Valor
Nombre	PROF-IPSEC-P5
Transform Set	TS-P5
IKEv2 Profile	PROF-P5

> En P2 (IKEv1 VTI) el `crypto ipsec profile` solo referenciaba el transform set. En P5 (IKEv2 VTI) se agrega `set ikev2-profile PROF-P5` para forzar el uso de IKEv2 en la negociación.

Interfaz Tunnel0 (VTI)

Parámetro	R1	R2
IP Tunnel	25.7.73.153/30	25.7.73.154/30
Tunnel source	Gi1	Gi2
Tunnel destination	25.7.73.134	25.7.73.129
Tunnel mode	ipsec ipv4	ipsec ipv4
Tunnel protection	PROF-IPSEC-P5	PROF-IPSEC-P5

Fase 2 — IPSec Transform Set

Parámetro	Valor
Cifrado ESP	AES 256
Integridad ESP	SHA-256 HMAC

Parámetro	Valor
Modo	Tunnel

Scripts de configuración

ISP

```
enable
configure terminal
hostname ISP

interface GigabitEthernet1
 ip address 25.7.73.130 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet2
 ip address 25.7.73.133 255.255.255.252
 no shutdown
 exit

end
write memory
```

R1

```
enable
configure terminal
hostname R1

interface GigabitEthernet1
 ip address 25.7.73.129 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.137 255.255.255.248
 no shutdown
 exit

ip route 0.0.0.0 0.0.0.0 25.7.73.130

! --- IKEv2 Proposal ---
crypto ikev2 proposal PROP-P5
 encryption aes-cbc-256
 integrity sha256
 group 14
 exit
```

```
! --- IKEv2 Policy ---
crypto ikev2 policy POL-P5
 proposal PROP-P5
 exit

! --- Keyring ---
crypto ikev2 keyring KEY-P5
 peer R2
  address 25.7.73.134
  pre-shared-key Cisco123!
  exit
 exit

! --- IKEv2 Profile ---
crypto ikev2 profile PROF-P5
 match identity remote address 25.7.73.134 255.255.255.255
 authentication local pre-share
 authentication remote pre-share
 keyring local KEY-P5
 exit

! --- Transform Set ---
crypto ipsec transform-set TS-P5 esp-aes 256 esp-sha256-hmac
 mode tunnel
 exit

! --- IPSec Profile referenciando IKEv2 (diferencia clave vs P2) ---
crypto ipsec profile PROF-IPSEC-P5
 set transform-set TS-P5
 set ikev2-profile PROF-P5
 exit

! --- Interfaz Tunnel VTI ---
interface Tunnel0
 ip address 25.7.73.153 255.255.255.252
 tunnel source GigabitEthernet1
 tunnel destination 25.7.73.134
 tunnel mode ipsec ipv4
 tunnel protection ipsec profile PROF-IPSEC-P5
 no shutdown
 exit

! --- Ruta hacia LAN-B a través del túnel ---
ip route 25.7.73.144 255.255.255.248 25.7.73.154

end
write memory
```

R2

```
enable
configure terminal
hostname R2

interface GigabitEthernet2
 ip address 25.7.73.134 255.255.255.252
 no shutdown
 exit

interface GigabitEthernet3
 ip address 25.7.73.145 255.255.255.248
 no shutdown
 exit

ip route 0.0.0.0 0.0.0.0 25.7.73.133

! --- IKEv2 Proposal ---
crypto ikev2 proposal PROP-P5
 encryption aes-cbc-256
 integrity sha256
 group 14
 exit

! --- IKEv2 Policy ---
crypto ikev2 policy POL-P5
 proposal PROP-P5
 exit

! --- Keyring ---
crypto ikev2 keyring KEY-P5
 peer R1
  address 25.7.73.129
  pre-shared-key Cisco123!
  exit
 exit

! --- IKEv2 Profile ---
crypto ikev2 profile PROF-P5
 match identity remote address 25.7.73.129 255.255.255.255
 authentication local pre-share
 authentication remote pre-share
 keyring local KEY-P5
 exit

! --- Transform Set ---
crypto ipsec transform-set TS-P5 esp-aes 256 esp-sha256-hmac
 mode tunnel
 exit

! --- IPSec Profile referenciando IKEv2 ---
crypto ipsec profile PROF-IPSEC-P5
 set transform-set TS-P5
```

```
 set ikev2-profile PROF-P5
 exit

! --- Interfaz Tunnel VTI ---
interface Tunnel0
 ip address 25.7.73.154 255.255.255.252
 tunnel source GigabitEthernet2
 tunnel destination 25.7.73.129
 tunnel mode ipsec ipv4
 tunnel protection ipsec profile PROF-IPSEC-P5
 no shutdown
 exit

! --- Ruta hacia LAN-A a través del túnel ---
ip route 25.7.73.136 255.255.255.248 25.7.73.153

end
write memory
```

SW-A y SW-B (IOSvL2)

```
enable
configure terminal

interface GigabitEthernet0/0
 switchport mode access
 switchport access vlan 1
 no shutdown
 exit

interface GigabitEthernet0/1
 switchport mode access
 switchport access vlan 1
 no shutdown
 exit

end
write memory
```

PC1 (VPCS)

```
ip 25.7.73.138 255.255.255.248 25.7.73.137
save
```

PC2 (VPCS)

```
ip 25.7.73.146 255.255.255.248 25.7.73.145
save
```

Verificación y pruebas

> Comandos ejecutados en R1 y R2. El ISP no participa de la VPN; los PCs solo ejecutan `ping`.

Comandos de verificación

```
show interface Tunnel0
show crypto ikev2 sa
show crypto ipsec sa
show ip route
```

Resultados obtenidos — R1

show interface Tunnel0:

```
Tunnel0 is up, line protocol is up
  Internet address is 25.7.73.153/30
  Tunnel protocol/transport IPSEC/IP
  Tunnel source 25.7.73.129 (GigabitEthernet1), destination 25.7.73.134
  Tunnel protection via IPSec (profile "PROF-IPSEC-P5")
```

show crypto ikev2 sa:

```
Tunnel-id Local                 Remote                fvrf/ivrf            Status
2         25.7.73.129/500       25.7.73.134/500       none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth
sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/616 sec
```

show crypto ipsec sa (fragmento):

```
local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
in use settings ={Tunnel, }
Status: ACTIVE(ACTIVE)
```

show ip route (fragmento):

```
S      25.7.73.144/29 [1/0] via 25.7.73.154
C      25.7.73.152/30 is directly connected, Tunnel0
L      25.7.73.153/32 is directly connected, Tunnel0
```

Pruebas de conectividad

Ping PC1 → PC2:

```
PC1> ping 25.7.73.146
25.7.73.146 icmp_seq=1 timeout
84 bytes from 25.7.73.146 icmp_seq=2 ttl=62 time=73.824 ms
84 bytes from 25.7.73.146 icmp_seq=3 ttl=62 time=67.862 ms
84 bytes from 25.7.73.146 icmp_seq=4 ttl=62 time=72.613 ms
```

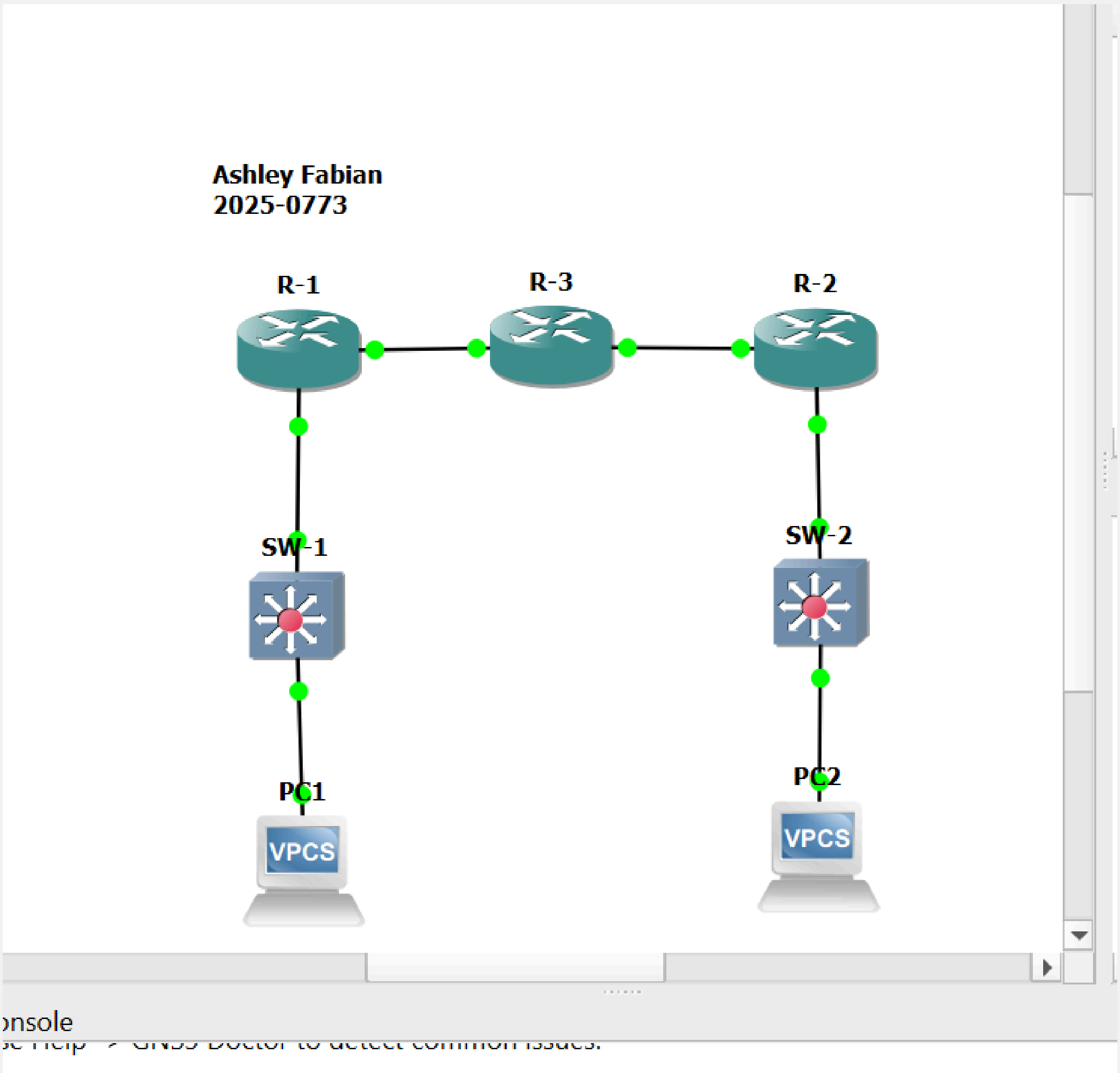
Ping PC2 → PC1:

```
PC2> ping 25.7.73.138
84 bytes from 25.7.73.138 icmp_seq=1 ttl=62 time=139.594 ms
84 bytes from 25.7.73.138 icmp_seq=2 ttl=62 time=82.685 ms
84 bytes from 25.7.73.138 icmp_seq=3 ttl=62 time=48.124 ms
```

> El primer paquete de PC1 dio timeout por la negociación inicial de IKEv2 (IKE_SA_INIT + IKE_AUTH). Una vez establecida la SA, los paquetes subsiguientes viajan cifrados sin interrupción. Este comportamiento es idéntico al observado en P4 y es completamente normal en VPNs basadas en políticas o VTI con negociación on-demand.

Capturas de pantalla

> Insertar aquí las capturas de:

> 1. Topología completa en GNS3 con nombre y matrícula visible
>
> 
>

2. `show interface Tunnel0` en R1 y R2 (IPSEC/IP, up/up, profile PROF-IPSEC-P5)

```
R1#show interface Tunnel0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 25.7.73.153/30
  MTU 9938 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 25.7.73.129 (GigabitEthernet1), destination 25.7.73.134
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with GigabitEthernet1
          Set of tunnels with source GigabitEthernet1, 1 member (includes iterators), on interfa
ce <OK>
  Tunnel protocol/transport IPSEC/IP
  Tunnel TTL 255
  Tunnel transport MTU 1438 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Tunnel protection via IPSec (profile "PROF-IPSEC-P5")
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:18:16
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     7 packets input, 588 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     7 packets output, 588 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
R1#
```

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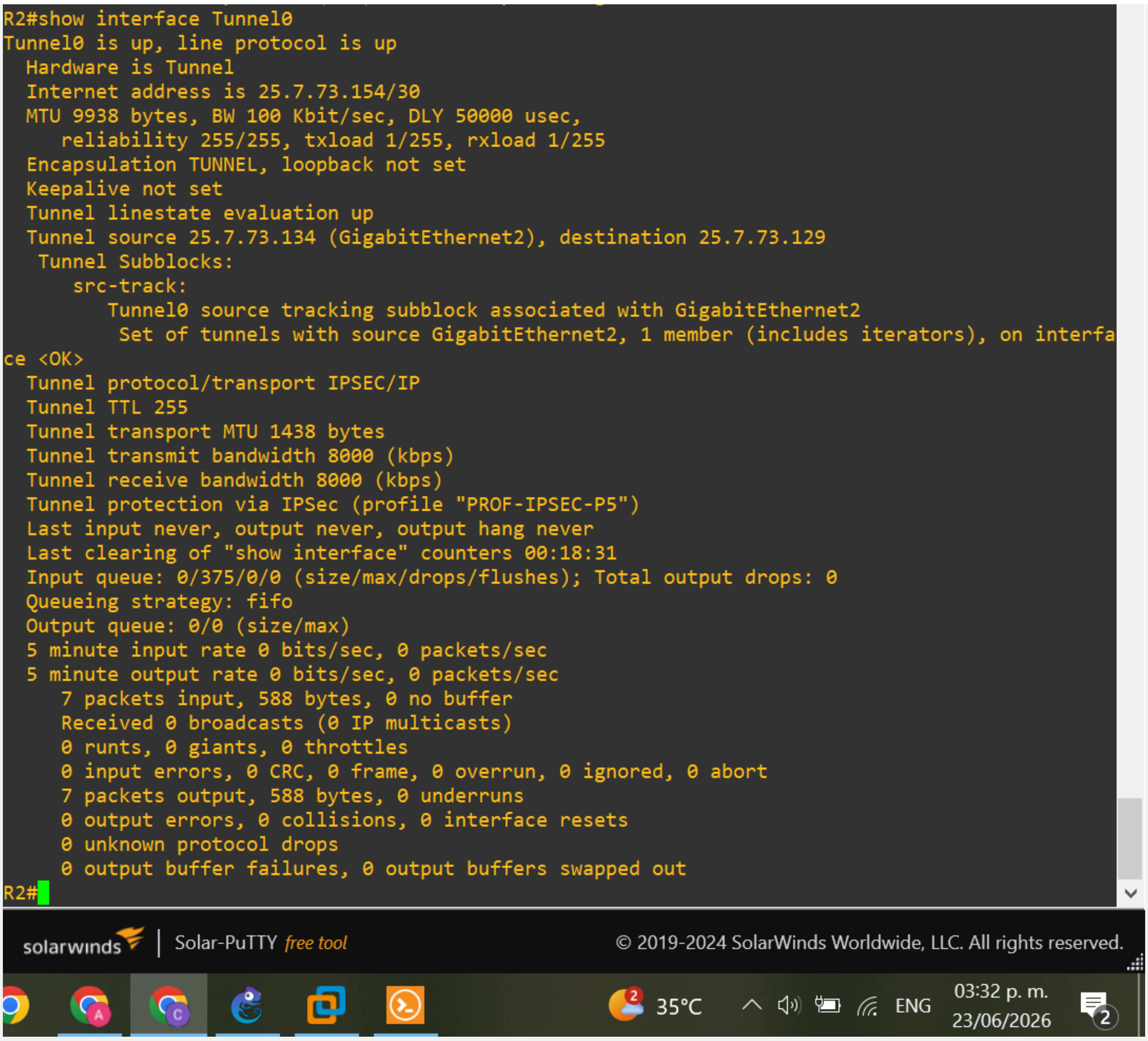
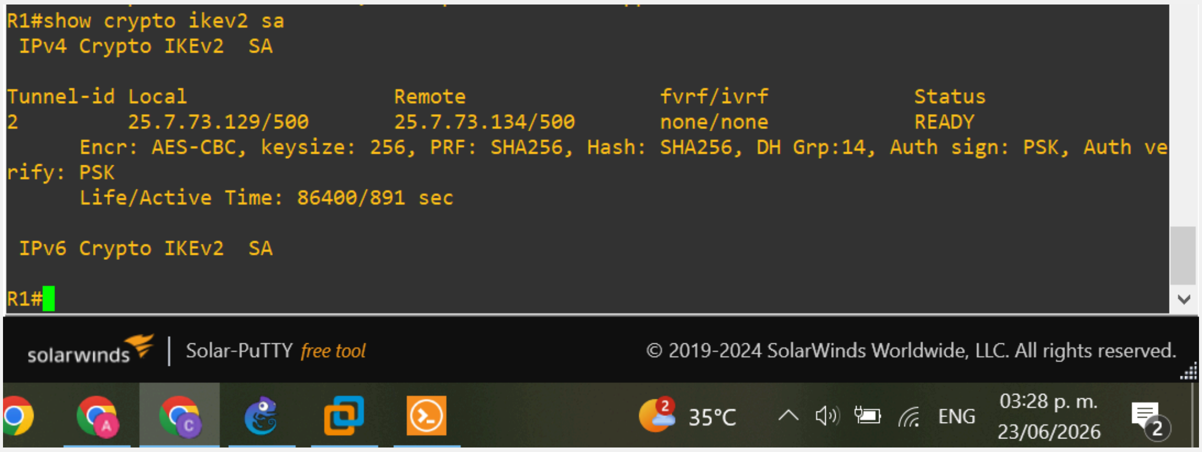
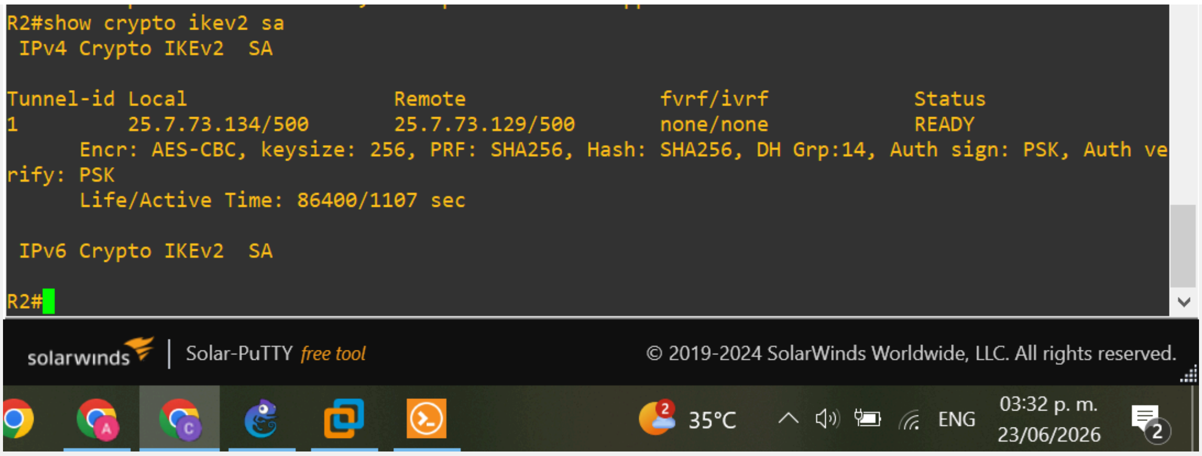


```
R2#show interface Tunnel0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 25.7.73.154/30
  MTU 9938 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 25.7.73.134 (GigabitEthernet2), destination 25.7.73.129
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with GigabitEthernet2
          Set of tunnels with source GigabitEthernet2, 1 member (includes iterators), on interfa
ce <OK>
  Tunnel protocol/transport IPSEC/IP
  Tunnel TTL 255
  Tunnel transport MTU 1438 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Tunnel protection via IPSec (profile "PROF-IPSEC-P5")
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:18:31
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     7 packets input, 588 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     7 packets output, 588 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
R2#
```

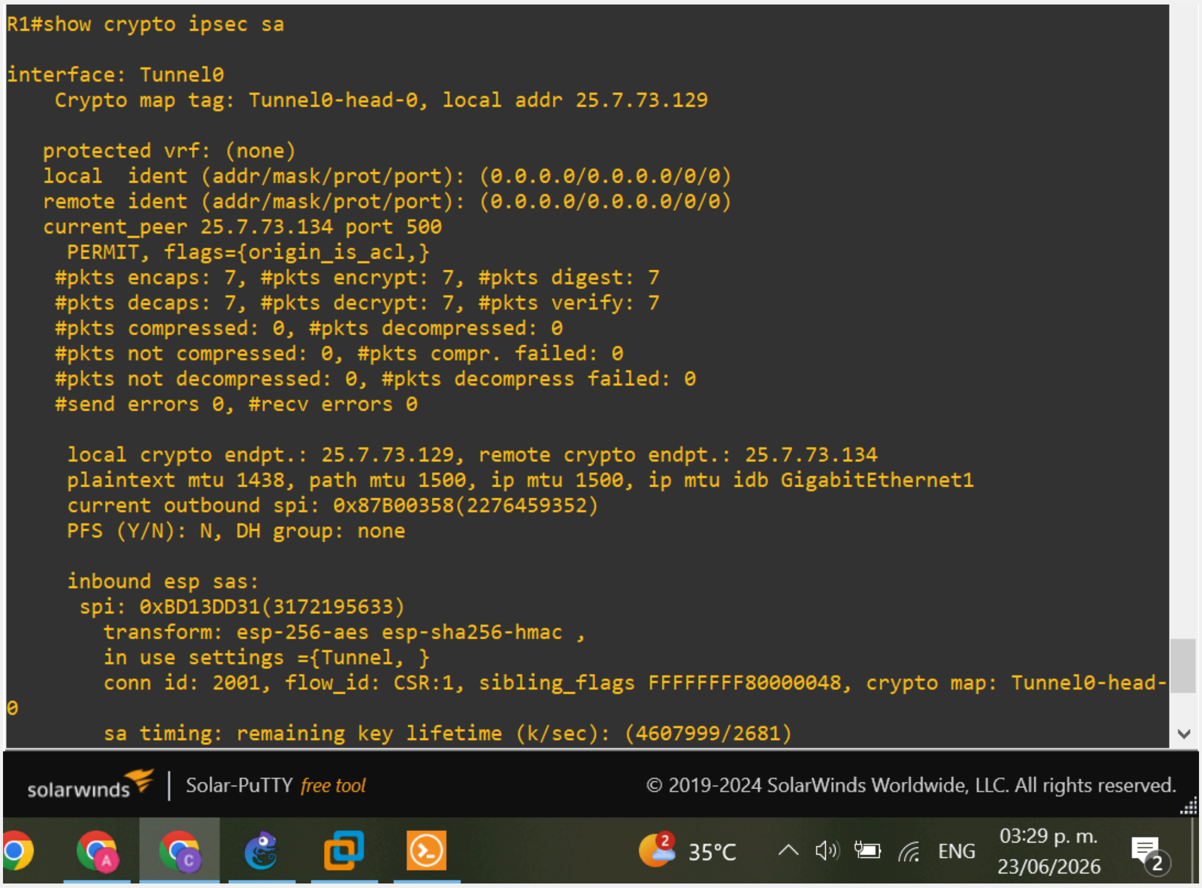

3. `show crypto ikev2 sa` en R1 y R2 (Status READY)

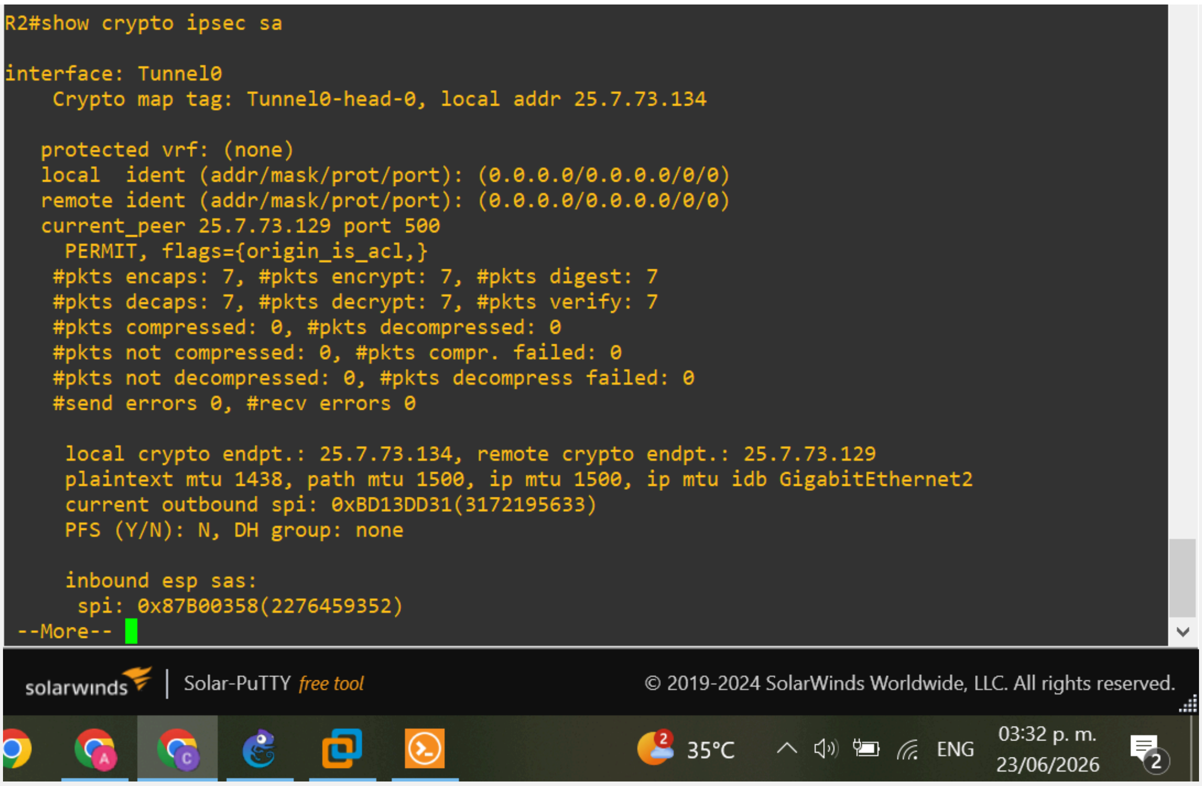


```
R1#show crypto ikev2 sa
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
2         25.7.73.129/500       25.7.73.134/500       none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth ve
rify: PSK
      Life/Active Time: 86400/891 sec

 IPv6 Crypto IKEv2  SA

R1#
```



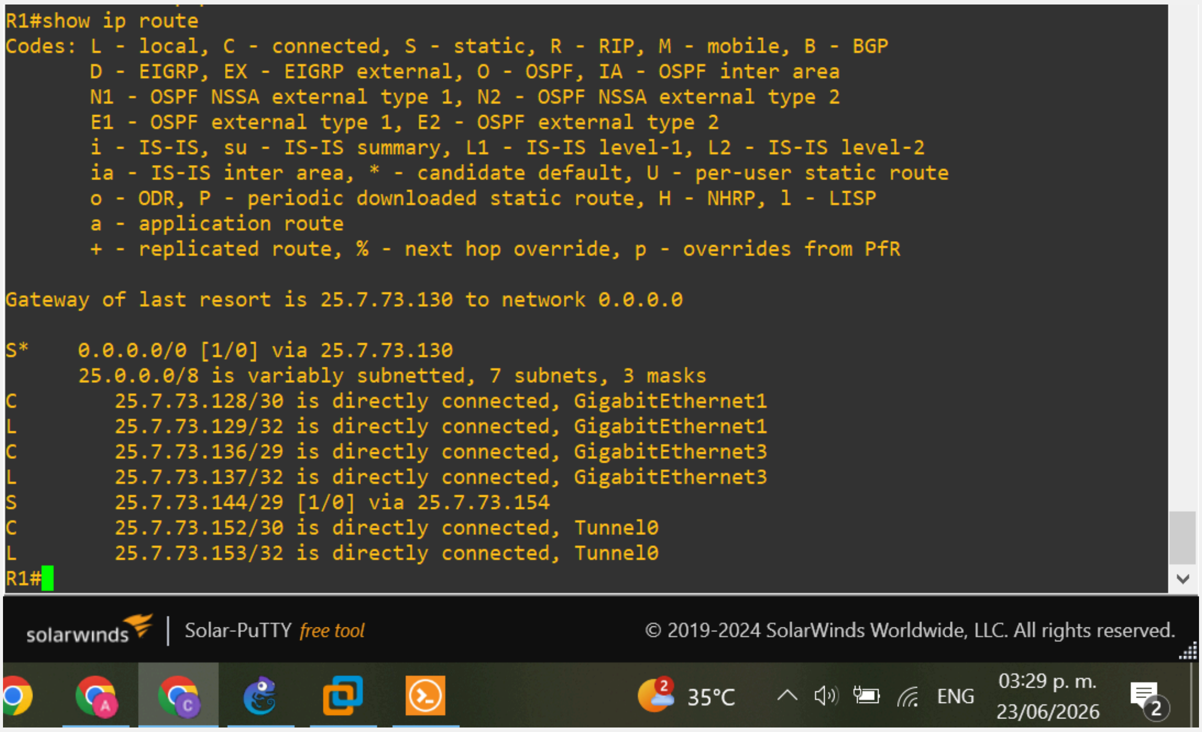
4. `show crypto ipsec sa` en R1 y R2 (Status ACTIVE, identifiers 0.0.0.0/0.0.0.0)





```
R2#show crypto ipsec sa

interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 25.7.73.134

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
   current_peer 25.7.73.129 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 7
    #pkts decaps: 7, #pkts decrypt: 7, #pkts verify: 7
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 25.7.73.134, remote crypto endpt.: 25.7.73.129
     plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet2
     current outbound spi: 0xBD13DD31(3172195633)
     PFS (Y/N): N, DH group: none

     inbound esp sas:
      spi: 0x87B00358(2276459352)
 --More--
```


5. `show ip route` en R1 y R2 (ruta estática via Tunnel0)



```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 25.7.73.130 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 25.7.73.130
      25.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C        25.7.73.128/30 is directly connected, GigabitEthernet1
L        25.7.73.129/32 is directly connected, GigabitEthernet1
C        25.7.73.136/29 is directly connected, GigabitEthernet3
L        25.7.73.137/32 is directly connected, GigabitEthernet3
S        25.7.73.144/29 [1/0] via 25.7.73.154
C        25.7.73.152/30 is directly connected, Tunnel0
L        25.7.73.153/32 is directly connected, Tunnel0
R1#
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 25.7.73.133 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 25.7.73.133
      25.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C        25.7.73.132/30 is directly connected, GigabitEthernet2
L        25.7.73.134/32 is directly connected, GigabitEthernet2
S        25.7.73.136/29 [1/0] via 25.7.73.153
C        25.7.73.144/29 is directly connected, GigabitEthernet3
L        25.7.73.145/32 is directly connected, GigabitEthernet3
C        25.7.73.152/30 is directly connected, Tunnel0
L        25.7.73.154/32 is directly connected, Tunnel0
R2#
```

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6. Ping PC1 → PC2 (mostrando primer timeout y éxitos posteriores)

```
PC1>  ping 25.7.73.146
25.7.73.146 icmp_seq=1 timeout
84 bytes from 25.7.73.146 icmp_seq=2 ttl=62 time=73.824 ms
84 bytes from 25.7.73.146 icmp_seq=3 ttl=62 time=67.862 ms
84 bytes from 25.7.73.146 icmp_seq=4 ttl=62 time=72.613 ms

PC1>
PC1>
```

7. Ping PC2 → PC1 (exitoso desde el primer paquete)

```
PC2> ping 25.7.73.138
84 bytes from 25.7.73.138 icmp_seq=1 ttl=62 time=139.594 ms
84 bytes from 25.7.73.138 icmp_seq=2 ttl=62 time=82.685 ms
84 bytes from 25.7.73.138 icmp_seq=3 ttl=62 time=48.124 ms

PC2>
PC2>
```

Conclusión

Se configuró exitosamente una VPN IPSec IKEv2 Site-to-Site basada en rutas (VTI) entre R1 y R2. La combinación de IKEv2 con VTI representa la arquitectura más moderna y recomendada para VPNs Site-to-Site en entornos Cisco IOS-XE: IKEv2 reduce los intercambios de negociación a 4 mensajes (vs 6-9 de IKEv1 Main Mode), mientras que VTI elimina la necesidad de ACLs para definir tráfico interesante, simplificando la configuración y facilitando la integración con protocolos de enrutamiento dinámico. La SA IKEv2 alcanzó el estado READY con AES-256, SHA-256 y DH Group 14, y la conectividad bidireccional entre LAN-A (25.7.73.136/29) y LAN-B (25.7.73.144/29) fue verificada exitosamente.